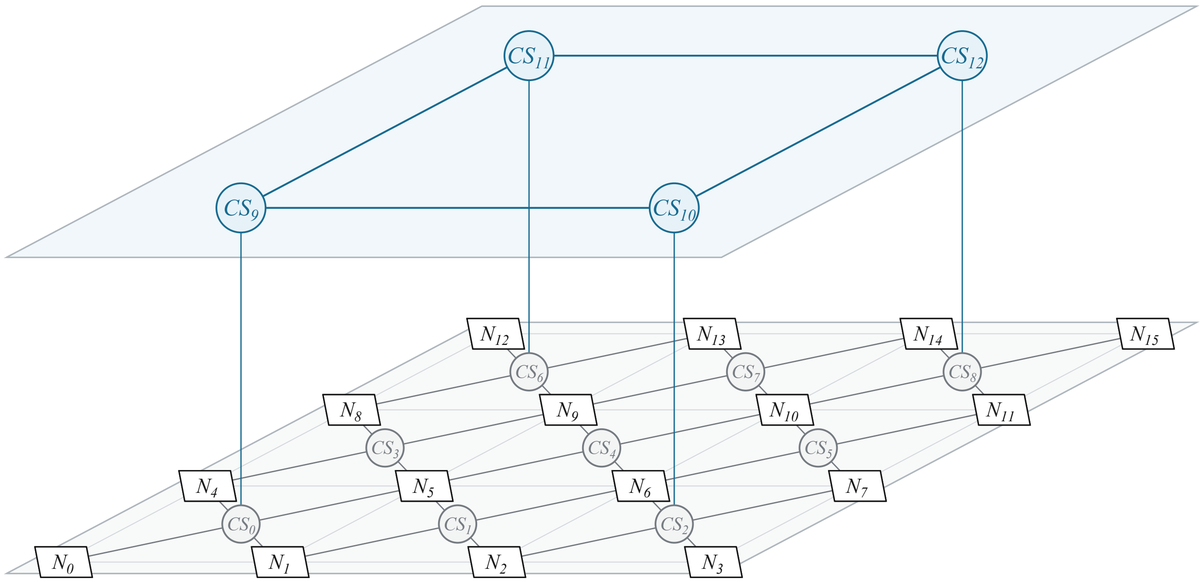
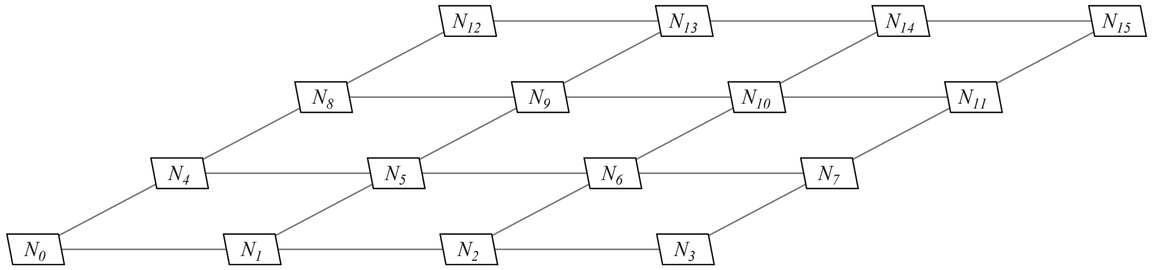
(a) Physical network and basic mesh

Nine lower CSs + four upper CSs

Each upper CS connects to its corresponding lower CS and two upper neighbors.



Basic 4 x 4 mesh



Lower CS — Upper CS

Node-to-CS links — Upper and inter-level links

No direct lower-CS-to-lower-CS links.

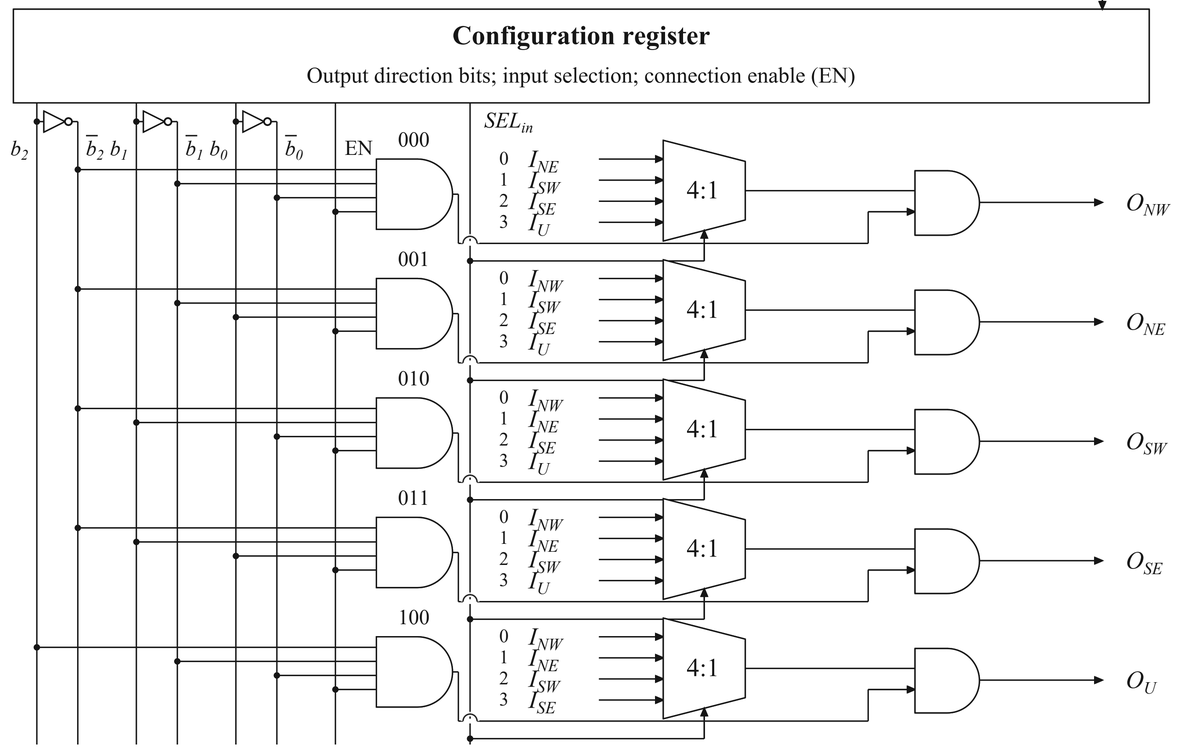
No direct upper-CS-to-node links. Links have separate transmit/receive wires.

Upper CS	Horizontal directions	Down to
CS_9	E, S	CS_0
CS_{10}	W, S	CS_2
CS_{11}	E, N	CS_6
CS_{12}	W, N	CS_8

(b) Single-connection configurable switches

Configuration from NI

Lower CS: five directions, five 4:1 MUXs



Direction code above each AND gate selects its output; only one gate can be enabled.

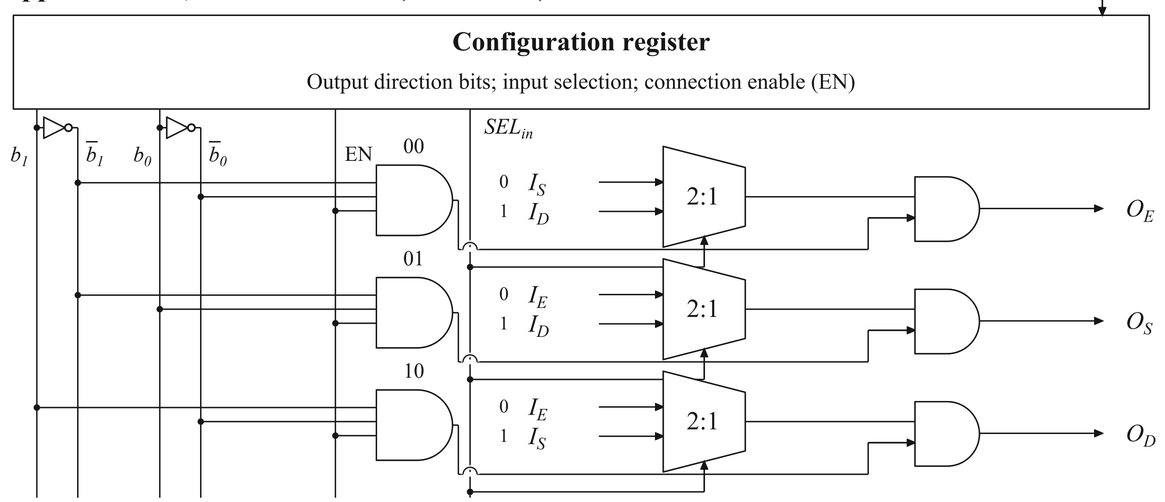
A bar means NOT; a dot means a junction. Input codes follow the listed order (0 at top).

NW / NE / SW / SE: corner nodes. U: the corresponding upper CS.

I / O: input / output. Same-direction inputs are excluded from each MUX.

Configuration from NI

Upper CS: east, south and down (CS9 shown)



Direction code above each AND gate selects its output; only one gate can be enabled.

A bar means NOT; a dot means a junction. Input codes follow the listed order (0 at top).

E / W / N / S: physical compass directions. D: down to the corresponding lower CS.